Matías Molina

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Data science - Machine learning - Statistics - Python - Scikit-learn - NumPy - Pandas - PyTorch - Seaborn - Jupyter - Git - Research - Teaching
Supervised learning - Unsupervised learning - Classification - Regression - Time series - Bayesian statistics - Neural networks - NLP - Vision

Education

2021 Ph.D. in Computer Science. (Intrinsic Phenomena in Zero-shot Classification)
AI> Machine learning> Transfer Learning > Zero-shot learning.
`Computer Vision, NLP, ANN, Bayesian Statistics.`

2012 Licentiate of Computer Science (equivalent to a MSc's degree)
2010 Analyst of Computer Science (equivalent to a BSc's degree)
Faculty of Mathematics, Astronomy, Physics and Computer Science (FAMAF).
National University of Córdoba, Argentina (UNC).

R&D

01.2021 - 12.2022 Data Science consultant (Freelance)
Data analysis and machine learning algorithms implementation: classification models, time series forecasting and data science tutoring.
`Python, Jupyter notebook, Git, SciKit-learn, Pandas, NumPy, Seaborn.`

06.2021 - 06.2022: Data Scientist - PreCog.finance (http://precog.finance)
Member of the R&D team. Research and prototyping of time series forecasting models for stock market analysis. Implementation and evaluation of different arbitrage strategies. Production development collaboration.
`Problem formalization, topic presentation for clients and investors. ETL pipeline, algorithms programming, backtesting. Python, Jupyter notebook, Git, SciKit-learn, Pandas, NumPy, Seaborn.`

04.2021 - 12.2021 Machine Learning Researcher - Industry collaboration (OVT).
Member of R&D team from the UNC consulting. Data exploration, machine learning research, anomaly detection, (train/val/test cycle), discussion of methodologies and result presentation to the clients.
`Classification techniques, word-embeddings (NLP), anomaly detection, sequential models, neural networks, SciKit-learn, Pandas, NumPy, Seaborn, Jupyter notebook, Git.`

2021 - 2021 Professor. Data Science Diploma at the Faculty of Economic Sciences (web).
Class preparation and presentation: fundamentals of Machine Learning, Impact of AI on economics and business, Exploratory data analysis. Machine learning models: Train-Validation-Test. Project assignment creation, team supporting and guidance (advisor).

2017, 2018 - Member of the teacher's assistants evaluation court at the UNC.
Management volunteer role. It includes evaluation of teacher's resumes, taking interviews and scoring.

2013 - Co-Advisor of MSc Thesis. Faculty of Mathematics, Astronomy, Physics and Computer Science.

2012-2015 - Python Developer. I've been developing in both industry (Harriague, IncluIT) and academia (FAMAF-UNC). Using Python, MongoDB and REST architecture.

Teaching

2014 - Present: Assistant Professor. CS degree - National University of Córdoba:
I've been teaching for many years at the University. During this period I was teaching different subjects and I was also earning more responsibilities.
`class material research and design, presentation, student advisoring, exam design, corrections.`

- 2020, 2021, 2022: Introduction to Algorithms.
- 2018, 2019: Introduction to Algorithms. Operative Systems.
- 2017: Paradigms of Programming Languages, Operative Systems
- 2016: Introduction to Algorithms. Operative Systems.
- 2015: Networks and Distributed Systems, Operative Systems.
- 2014: Computer Organization, Operative Systems.
- 2009: Undergraduate Teaching Assistant.

Publications

Zero-shot learning with partial attributes, M.Molina and J. Sánchez, ***ISICS*** *2018* [+]
Performance variability in zero-shot classification. M.Molina and J. Sánchez, ***NeurIPS*** *2020* [+]
Trading-off information modalities for zero-shot classification. M.Molina and J. Sánchez, ***WACV*** *2022* [+]
Synthetic dataset generation for ZS-Classification evaluation (Sent to ***PRLetters****)*

Conferences

2020 - Presenter at LXAI workshop at NIPS.
2020 - Attendance at BMVC, CVPR, NIPS.
2020 - Attendance at MachinesCanSee 2020 http://machinescansee.com/.
2018 - Presenter at the International Symposium on Intelligent Computing Systems 2018, Mérida, México. Zero-Shot Learning with Partial Attributes.
2017 Attendance to the Workshop Hands On Data, Catholic University of Córdoba, Argentina.
2017 Attendance to the XLIII CLEI – 46 JAIIO, Argentinian School of Artificial Intelligence. UTN, UNC. Córdoba, Argentina.
2017 Attendance to the Workshop Publishing your research. Faculty Chemistry Science, UNC, Argentina.

Scholarships

2015 - PhD scholarship provided by the National Council of Technical and Scientific Research (CONICET).
2011 - TICs Scholarship provided by The National Ministry of Education for undergraduate students

Languages

Spanish (native) – English (professional) – Portuguese (basic)